# Karan Panwar

Cambridge, UK | +44 7823 706 438 | karanpanwar8595@gmail.com
Portfolio | GitHub | LinkedIn

## Professional Summary

Data Scientist and AI Engineer holding an MSc in Computer Science (Distinction) with specialized expertise in Large Language Models (LLMs), Generative AI architecture, predictive modeling, statistical analysis, and production microservices. First author of a peer-reviewed Springer publication on scalable LLM-driven intervention systems. Proven track record across end-to-end data science and AI lifecycles—ranging from dataset curation, exploratory data analysis (EDA), and feature engineering to model benchmarking, RAG pipelines, and deploying low-latency microservices using Python, PyTorch, LangChain, FastAPI, and AWS.

## Core Technical Skills

**AI Engineering & GenAI:** Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), LangChain, Hugging Face Transformers, Vector Embeddings, Google Genkit
**Data Science & Machine Learning:** PyTorch, TensorFlow, Scikit-Learn, XGBoost, CNNs, LSTMs, Statistical Modeling, Time-Series Forecasting, Exploratory Data Analysis (EDA), Feature Engineering
**Quantum Machine Learning:** Qiskit, Variational Quantum Circuits (VQC), Quantum-Classical Hybrid Neural Networks, Custom Simulator Benchmarking
**Programming & Databases:** Python, NumPy, Pandas, SQL, TypeScript, Java, Django, FastAPI, Flask, MySQL, MongoDB, SQLite, Structured JSON Analytics
**DevOps & Software Tools:** AWS (Certified Cloud Practitioner), Docker, Git, REST API Architecture, Asynchronous Workflows, LaTeX

## Research and Professional Experience

**Research Assistant (Generative AI & Data Science) – Klip Global** Sep 2025 – Jun 2026
*London, UK (Remote from Cambridge)*

- Architected core conversational AI microservices and personalized recommendation workflows for KAI Navigator, an AI-powered digital wellbeing platform.
- Processed and structured 700 complex JSON assessment records for behavioral trend analysis, data cleaning, feature design, and platform integration.
- Engineered a hybrid recommendation engine combining Transformer-based NLP semantic embeddings with unsupervised clustering algorithms.
- Conducted quantitative model benchmarking, evaluating recommendation quality against a collaborative-filtering baseline to achieve optimal cluster separation and user relevance.
- Integrated backend Python microservices (LangChain, Hugging Face) with a Django REST framework and Next.js frontend, ensuring low-latency communication.
- First-authored a peer-reviewed paper describing a scalable LLM-based system for early intervention and workforce resilience, published in Springer's *Artificial Intelligence in Healthcare* (AIiH 2026) proceedings.

**Research Assistant – Quantum Neural Networks** Jan 2025 – Apr 2025
*Anglia Ruskin University, Cambridge, UK*

- Investigated quantum machine learning for image classification by implementing a hybrid quantum-classical neural network inside a custom PyTorch simulator.
- Designed a 10-qubit circuit architecture featuring 4 variational layers, Ry and Rz rotation gates, and controlled-X entanglement linked to a classical linear classification layer.
- Conducted experimental comparative evaluation against a classical CNN baseline across a curated dataset of 3,200 real and synthetic face images.
- Analyzed quantitative performance trade-offs across test accuracy, loss convergence, training execution time, and model parameter complexity.
- Recorded 86.45% test accuracy for the quantum hybrid model (8,202 trainable parameters) versus 91.8% for the classical CNN (1,345,377 trainable parameters), demonstrating parameter efficiency.

## Selected AI & Data Science Projects

**SmartStock – AI Financial Forecast** *Python, LSTM, FastAPI, Pandas, Docker*

- Developed a financial forecasting dashboard utilizing Long Short-Term Memory (LSTM) recurrent networks for multi-step time-series prediction of market trends.
- Built automated data ingestion pipelines connecting to external market-data APIs, applying rolling window transformations, feature scaling, and missing-value imputation.
- Containerized asynchronous FastAPI endpoints using Docker to serve real-time predictions and analytical metrics with low-latency execution.

**Predictive Maintenance Platform – Turbofan Failure Diagnostics** *Python, XGBoost, SHAP, FastAPI,*

*Streamlit*

- Built an end-to-end predictive maintenance system modeling sensor degradation using NASA's C-MAPSS turbofan engine dataset.
- Engineered multi-sensor time-domain features and trained gradient-boosted decision trees (XGBoost) alongside deep sequence models to forecast Remaining Useful Life (RUL).
- Incorporated SHAP (SHapley Additive exPlanations) values to provide model explainability for critical maintenance decision-making.
- Deployed an interactive Streamlit dashboard backed by a RESTful FastAPI service packaged in Docker containers.

**EmotionSense – Multi-Modal Audio-Visual Sentiment Classifier** *Python, OpenCV, TensorFlow, Flask*

- Developed a multi-modal sentiment recognition prototype extracting visual facial features via CNNs and acoustic descriptors using Mel-Frequency Cepstral Coefficients (MFCCs).
- Evaluated model performance across 7 emotional categories, achieving over 88% multi-class classification accuracy on cross-validated held-out project test data.
- Wrapped the inference pipeline into a lightweight web interface powered by Flask for real-time video stream testing.

**KpAgent – Autonomous AI Career Assistant Platform** *Next.js, TypeScript, NLP, Python, Vector Search*

- Built an intelligent career management application offering automated CV parsing, personalized job recommendation matching, and role-specific interview simulation.
- Implemented NLP text processing pipelines and semantic vector similarity scoring to align user profiles with structured job specifications.

## Peer-Reviewed Publication

**Panwar, K.**, Cirstea, S., Kalidindi, S., & Chakraborty, A. (2026). "A Scalable LLM-Based System for Early Mental Health Intervention and Workforce Resilience Using Genkit." *Artificial Intelligence in Healthcare (AIiH 2026)*, Lecture Notes in Computer Science (LNCS), vol. 16877, pp. 278–291. Springer, Cham.
DOI: 10.1007/978-3-032-35393-1_21

## Education

**MSc Computer Science (Distinction)** Sep 2024 – Dec 2025
*Anglia Ruskin University, Cambridge, UK*

- *Relevant Modules:* Machine Learning in Finance, AI for Imaging, Semantic Web Technologies, Advanced Web Solutions, Cybersecurity and AI.

**BSc Computer Applications (First Class Honors)** May 2021 – May 2024
*St. Xavier's College, Ahmedabad, India*

- *Relevant Modules:* Big Data Analytics, Statistical Modelling, Data Structures, Python Programming, Database Management.

## Analytical Methodologies & Certifications

**Data Science & Engineering Methodologies:** Experimental Design, Hypothesis Testing, Model Evaluation (RMSE, MAPE, Precision/Recall, F1-Score, ROC-AUC), Hyperparameter Tuning, Cross-Validation, RAG Architecture, Agile Development

**Certifications & Honors:** AWS Certified Cloud Practitioner | IBM Data Science Professional Certificate | AI & ML using Python (IIT Madras / SWAYAM, 86%) | Quantum Computing Fundamentals (IBM Qiskit) | Advanced Deep Learning (Coursera)